

BEAUFORT, NC, USA

WMO: 723037

Lat: 34.733N

Lon: 76.657W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 93765

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-1.9	-12.9	1.2	-1.5	-10.5	1.5	0.6	12.4	7.1	11.5	6.0	4.6	0	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	31.3	25.8	30.5	25.6	29.8	25.3	27.4	29.7	26.9	29.2	26.4	28.7	5.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.2	28.8	26.2	21.6	28.5	25.7	20.9	28.3	86.3	29.9	84.1	29.3	82.0	28.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.4	8.5	DB	-7.1	33.6	2.5	2.0	-8.9	35.1	-10.4	36.3	-11.8	37.5	-13.6	38.9	
			WB	-8.6	28.1	2.1	0.6	-10.2	28.6	-11.4	28.9	-12.6	29.3	-14.2	29.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	7.5	8.7	12.0	16.8	21.3	25.3	26.9	26.4	24.0	19.1	13.5	10.4
	DBStd	7.80	5.39	4.57	4.45	3.68	3.09	2.13	1.70	1.74	2.35	4.13	4.39	4.89
	HDD10.0	292	113	72	31	2	0	0	0	0	0	1	16	57
	HDD18.3	1330	335	269	199	70	11	0	0	0	1	42	153	249
	CDD10.0	3107	37	37	93	206	349	460	523	509	421	282	122	68
	CDD18.3	1103	0	1	3	24	102	210	265	251	171	65	9	2
	CDH23.3	9099	0	0	4	36	445	1813	2777	2483	1201	322	19	0
	CDH26.7	2189	0	0	1	1	41	380	816	700	221	28	1	0
Wind	WSAvg	4.2	4.3	4.5	4.7	4.9	4.5	4.4	4.4	3.6	3.8	3.8	3.9	4.0
Precipitation	PrecAvg	1468	124	106	105	81	121	119	174	173	155	107	95	105
	PrecMax	2324	243	314	288	237	341	278	368	393	401	331	270	292
	PrecMin	890	36	34	8	8	28	7	44	36	8	5	23	29
	PrecStd	279	58	62	54	51	72	62	75	95	96	79	57	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	21.8	23.4	26.0	29.3	32.0	32.6	32.7	30.5	28.9	25.4	22.3
		MCWB	18.0	18.0	18.3	21.6	23.7	24.8	26.3	26.0	24.2	24.6	20.6	20.2
	2%	DB	19.0	19.2	21.7	24.3	27.6	30.6	31.3	31.2	29.5	27.4	23.2	21.0
		MCWB	16.9	16.1	17.3	20.1	23.3	25.0	26.4	25.9	24.2	23.6	20.5	19.0
	5%	DB	17.7	17.7	20.2	23.2	26.7	29.5	30.5	30.3	28.8	26.3	22.2	19.7
		MCWB	16.0	15.2	16.6	19.5	22.6	24.7	26.0	25.6	24.0	22.7	19.4	17.9
	10%	DB	16.2	16.2	18.9	22.3	25.8	28.6	29.8	29.6	27.9	25.2	20.9	18.4
		MCWB	14.4	14.0	16.0	18.8	22.0	24.4	25.7	25.3	23.8	21.8	18.2	16.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	19.0	19.6	23.0	25.1	27.3	28.0	28.0	26.7	25.9	22.8	21.0
		MCDB	19.8	21.0	21.6	24.5	27.9	29.4	30.7	30.3	28.6	27.2	24.3	21.8
	2%	WB	17.5	17.4	18.6	21.5	24.2	26.6	27.5	27.3	25.8	24.6	21.6	19.5
		MCDB	18.5	18.5	20.7	23.4	26.8	28.9	30.0	29.6	28.0	26.5	22.7	20.3
	5%	WB	16.2	15.8	17.6	20.5	23.5	25.9	27.0	26.8	25.3	23.6	19.9	18.3
		MCDB	17.6	17.2	19.5	22.3	25.9	28.4	29.4	29.1	27.7	25.8	21.4	19.3
	10%	WB	14.7	14.3	16.6	19.4	22.8	25.4	26.6	26.2	24.7	22.7	18.6	16.8
		MCDB	16.1	16.0	18.4	21.4	25.0	27.9	29.0	28.6	27.1	24.7	20.1	18.2
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.9	8.8	7.8	6.8	6.0	5.6	6.0	6.6	8.0	9.4	8.9
		MCDBR	9.8	9.7	9.3	7.4	6.6	6.4	6.0	6.3	6.8	7.2	9.4	8.5
		MCWBR	8.3	7.6	6.2	4.7	3.6	3.5	3.4	3.3	3.5	4.5	6.7	7.1
	5% WB	MCDBR	9.1	9.0	7.7	6.4	5.5	5.4	5.1	5.3	5.5	6.2	8.5	7.7
		MCWBR	8.2	8.2	6.0	4.8	3.4	3.4	3.4	3.4	3.4	4.6	7.3	7.1
Clear-Sky Solar Irradiance	taub	0.327	0.336	0.362	0.401	0.445	0.490	0.515	0.499	0.424	0.373	0.348	0.336	
	taud	2.540	2.509	2.455	2.345	2.266	2.186	2.126	2.167	2.383	2.499	2.518	2.536	
	Ebn at Noon	882	911	911	886	847	806	782	788	837	858	848	847	
	Edh at Noon	81	93	106	124	136	147	155	147	113	93	82	78	
All-Sky Solar Radiation	RadAvg	2.57	3.31	4.38	5.56	6.17	6.37	6.01	5.47	4.52	3.84	2.85	2.29	
	RadStd	0.15	0.32	0.33	0.39	0.41	0.35	0.37	0.32	0.30	0.38	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+117

Nomenclature: See separate page